

Dear Path Robotics team,

I'm applying for the Senior Software Engineer, Perception Platform role because I've spent much of my career at exactly the boundary you describe: giving computer vision and ML researchers the infrastructure to turn experiments into reliable production systems.

I co-founded Cambrian and led its technical work across computer vision, ML, GPU computing, and real-time systems. We built proprietary perception technology that ran across mobile, web, cloud, and automotive applications, was licensed to Fortune 500 companies, and reached millions of users.

For our cloud perception stack, I built production Python/TensorFlow image-processing services that decomposed uploaded scenes through multiple CV and ML stages, including semantic structure, surface normals, geometry, lighting, and camera estimation, then used those representations to reconstruct and re-render the scene. The surrounding platform included APIs, S3-backed data flow, Docker CPU/GPU deployment, testing and model utilities, and elastically scaling AWS inference.

On-device, the systems problem was different but related. Segmentation and surface-normal CNNs could take 1-3 seconds on the hardware of the time, while the scene still had to evolve at frame rate. I built a priority-scheduled, backpressured C++ perception runtime that combined asynchronous learned results with tracking, geometry, temporal evidence, and sensor input without allowing the slowest estimator to stall the world model. That work required careful concurrency, memory management, SIMD, GPU/GPGPU optimization, model compression, and cross-platform deployment.

I've continued working on the infrastructure beneath ML systems. My current projects span transformer architecture research in Python and C++, a large Rust AI system where I work on inference-runtime internals, CUDA/Metal placement, GPU memory constraints, scheduling, telemetry, distributed compute, and multiple AI workloads sharing limited hardware.

What interests me about Path is that the Perception Platform team isn't isolated from either side of the problem. You need to understand what the researchers are doing, but the job is to build the architecture, interfaces, deployment machinery, observability, and developer tooling that let their work survive contact with a production fleet. That is work I know well and enjoy.

I'd be excited to bring that experience to the perception systems behind Path's robots.

Joel Teply github.com/CambrianTech